LAB # 9

RECURSIONS IN C++

Lab Tasks

1- Write a recursive function to obtain the first 25 numbers of a Fibonacci sequence. In a Fibonacci sequence the sum of two successive terms gives the third term. Following are the first few terms of the Fibonacci sequence: 1 1 2 3 5 8 13 21 34 55 89...

2- Develop a program that calculates the factorial of a number using recursion.

3- Develop a program that takes a word as input from the user terminated by enter key and displays the entered word in reverse order.

Input: recursion
Output: noisrucer

4- Write a program to generate and display the sum of first 10 terms of the following series using recursion:
(2 * 1) + 2, (2 * 2) + 2, (2 * 3) + 2 . . . . . .

1- Write a recursive function to obtain the first 25 numbers of a Fibonacci sequence. In a Fibonacci sequence the sum of two successive terms gives the third term. Following are the first few terms of the Fibonacci sequence: 1 1 2 3 5 8 13 21 34 55 89...

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```
#include <iostream>
using namespace std;

int fib(int a) {
    if (a == 0 || a == 1) {
        return a;
    }
    else {
        return fib(a - 1) + fib(a - 2);
    }
}

int main() {
    cout << "Muhammad Ali\n2025F-BIT-023" << endl;
    cout << "First 25 numbers of a Fibonacci Sequence: \n";
    for (int i = 0; i <= 25; i++) {
        cout << "Fibonacci of " << i << " = " << fib(i) << endl;
    }
    return 0;
}
```

```
Muhammad Ali
2025F-BIT-023
First 25 numbers of a Fibonacci Sequence:
Fibonacci of 0 = 0
Fibonacci of 1 = 1
Fibonacci of 2 = 1
Fibonacci of 3 = 2
Fibonacci of 4 = 3
Fibonacci of 5 = 5
Fibonacci of 6 = 8
Fibonacci of 7 = 13
Fibonacci of 8 = 21
Fibonacci of 9 = 34
Fibonacci of 10 = 55
Fibonacci of 11 = 89
Fibonacci of 12 = 144
Fibonacci of 13 = 233
Fibonacci of 14 = 377
Fibonacci of 15 = 610
Fibonacci of 16 = 987
Fibonacci of 17 = 1597
Fibonacci of 18 = 2584
Fibonacci of 19 = 4181
Fibonacci of 20 = 6765
Fibonacci of 21 = 10946
Fibonacci of 22 = 17711
Fibonacci of 23 = 28657
Fibonacci of 24 = 46368
Fibonacci of 25 = 75025
```

2- Develop a program that calculates the factorial of a number using recursion.

```cpp
#include <iostream>
using namespace std;

int fact(int a) {
    if (a < 0) {
        return -1;
    }
    if (a == 0) {
        return 1;
    }
    else {
        return (a*fact(a-1));
    }
}

int main() {
    cout << "Muhammad Ali\n2025F-BIT-023" << endl;
    int num;
    cout << "Enter number for factorial: ";
    cin >> num;
    cout << "The factorial of given number is: " << fact(num);
    return 0;
}
```

```
Muhammad Ali
2025F-BIT-023
Enter number for factorial: 5
The factorial of given number is: 120
```

3- Develop a program that takes a word as input from the user terminated by enter key and displays the entered word in reverse order.

Input: recursion
Output: noisrucer

onsoleApplication7 (Global Scope)

```cpp
#include <iostream>
using namespace std;

void reversed(string word) {
    if (word.empty()) {
        return;
    }
    else {
        reversed(word.substr(1));
        cout << word[0];
    }
}
int main() {
    cout << "Muhammad Ali\n2025F-BIT-023" << endl;
    string original;
    cout << "Enter word: ";
    cin >> original;
    cout << "Reversed: ";
    reversed(original);
    return 0;
}
```

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```
Muhammad Ali
2025F-BIT-023
Enter word: recursion
Reversed: noisrucer
```

4- Write a program to generate and display the sum of first 10 terms of the following series using recursion:

(2 * 1) + 2, (2 * 2) + 2, (2 * 3) + 2 . . . . . .

soleApplication7 (Global Scope)

```cpp
#include <iostream>
using namespace std;

int series(int n) {
    if (n == 1) {
        return (2 * 1) + 2;
    }
    return (2 * n) + 2 + series(n - 1);
}

int main() {
    cout<<"Muhammad Ali\n2025F-BIT-023" <<endl;
    int num = 10;
    int totalSum = series(num);
    cout << "The sum of first " << num << " terms is: " << totalSum << endl;
    return 0;
}
```

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```
Muhammad Ali
2025F-BIT-023
The sum of first 10 terms is: 130
```